IS3005 – Statistics in Practice I

Case Study I

A STATISTICAL ANALYSIS OF CUSTOMER CHURN IN THE SPEEDYCALL TELECOMMUNICATION COMPANY

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Abstract

Customer churn is a significant issue in the telecommunications industry. Keeping current customers is cheaper than gaining new ones. This study examines customer churn behavior at SpeedyCall, a telecommunications company in the USA. A sample of the company's customer data was used to look into the main demographic, behavioral, and service-related factors that affect churn. Data preprocessing included addressing missing values and choosing relevant variables for analysis. Descriptive statistics and visualizations helped spot differences between customers who left and those who stayed, while a logistic regression model was created to predict the chances of churn. The analysis showed that factors like contract type, monthly charges, and tenure have a strong impact on customer churn. The model had good predictive accuracy and identified customers who were likely to leave the company. The results of this study can help SpeedyCall create effective retention strategies and boost customer satisfaction.

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Introduction

Background of the Study

Customer churn refers to customers stopping their services and moving to competitors. This has become a big challenge in the telecommunications industry. With more competition and similar services among providers, companies like SpeedyCall struggle to keep their customers. Reducing churn is important because getting new customers costs much more than keeping the ones they already have. By understanding the reasons why customers leave, both behavioral and service-related, SpeedyCall can make better decisions to improve customer satisfaction and loyalty.

Objectives

- To identify the key demographic, behavioral, and service-related factors influencing customer churn among SpeedyCall customers.
- To compare the profiles of churned and non-churned customers to understand their differences.
- To provide practical recommendations to help *SpeedyCall* minimize churn and enhance customer retention strategies.
- To develop a statistical model that predicts the likelihood of customer churn based on selected variables.

- **Significance of the study**
  This study is significant as it provides valuable insights into customer behavior within the telecommunication sector. By identifying the major causes of churn, *SpeedyCall* can focus on high-risk customer segments and improve its service quality, pricing strategies, and customer relationship management. The findings will support the company's efforts to reduce revenue loss, strengthen customer loyalty, and maintain a competitive advantage in the market.

Data

The dataset used for this study, titled *SpeedyCall Customer Churn Data*, contains information about customers of a telecommunication company in the United States. It includes various demographic, behavioral, and service-related variables that describe customer characteristics and their service usage patterns. The main target variable in this study is **"Churn"**, which indicates whether a customer has discontinued the company's service (Yes) or remained with the company (No).For this analysis, a **subset** of the dataset was selected, including the variables most relevant for understanding customer behavior and predicting churn. The selected variables are summarized below.

Table 1: Variable Description

Variable Name	Description	Type
gender	Indicates whether the customer is Male or Female	Categorical
SeniorCitizen	Shows if the customer is a senior citizen (1 = Yes, 0 = No)	Categorical
Partner	Indicates if the customer has a partner(Yes/No)	Categorical
Dependents	Indicates if the customer has dependents (children, family) (Yes/No)	Categorical
Tenure	Number of months the customer has stayed with the company	Numerical
PhoneService	Whether the customer has a phone service(Yes/No)	Categorical
InternetService	Type of internet connection (DSL, Fiber optic, or No)	Categorical
Contract	Type of contract (Month-to-month, One year, Two year)	Categorical

MonthlyCharges	Monthly payment made by the customer (in USD)	Numerical
TotalCharges	Total amount charged during the customer's tenure	Numerical
Churn	Indicates whether the customer has left (Yes) or stayed (No)	Categorical (Target Variable)

Data Preprocessing:

Prior to analysis, data cleaning and preprocessing steps were performed. Missing or inconsistent data were identified and treated by imputation or removal where appropriate. Categorical variables such as *Gender*, *Partner*, and *Contract* were converted into factor variables for analysis. Continuous variables such as *Tenure*, *MonthlyCharges*, and *TotalCharges* were checked for outliers and normalized when required. The cleaned subset was then used for both descriptive and predictive analyses.

Data Analysis

OBJECTIVE 1 : To identify key demographic, behavioral, and service-related factors influencing customer churn among SpeedyCall customers.

01.Contract Type vs Churn

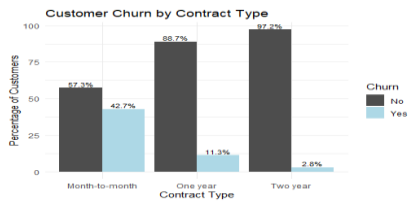


Figure 1. Contract Type vs Churn

Customers on a **Month-to-month** contract have the highest churn rate at **42.7%**. In stark contrast, customers with **One year** and especially **Two year** contracts show significantly lower churn rates, at 11.3% and 2.8%, respectively.

02. InternetService Type vs Churn

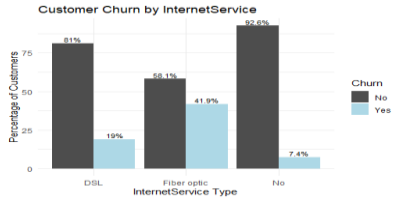


Figure 2. InternetService Type vs Churn

Customers using **Fiber optic** service have the highest churn rate at **41.9%**. Customers with **DSL** have a much lower churn rate (19%), and those with **No** internet service have the lowest churn (7.4%).

03. PhoneService Type vs Churn

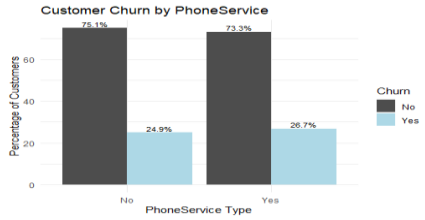


Figure 3. PhoneService Type vs Churn

Customers with and without phone service show **similar churn rates** (26.7% vs. 24.9%, respectively). This suggests that the presence of phone service alone does not have a strong differentiating impact on churn.

04. SeniorCitizen Type vs Churn

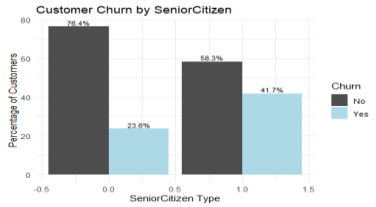


Figure 4. SeniorCitizen *Type vs Churn*

Senior Citizens (1.0) have a much higher churn rate at 41.7%. This is nearly double the churn rate of non-senior citizens (23.6%)

05.Dependents Type vs Churn

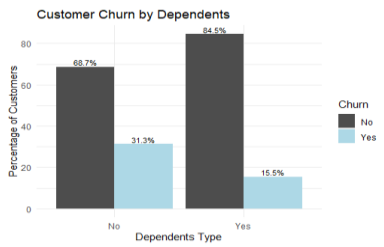


Figure 5. Dependents Type vs Churn

Customers **without dependents (No)** have a significantly **higher churn rate** (31.3%) compared to those with dependents (15.5%).

06.Partner vs Churn

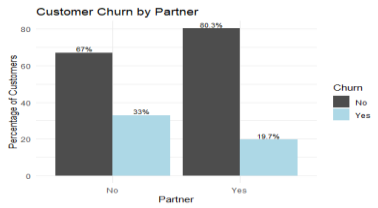


Figure 6. Partner Type vs Churn

Customers **without a partner (No)** have a higher churn rate at **33%**. The churn rate for customers with a partner is lower, at 19.7%.

OBJECTIVE 2 : To compare churned vs non-churned customer profiles.

<u>Numerical Variables</u>

01. Monthly Charges

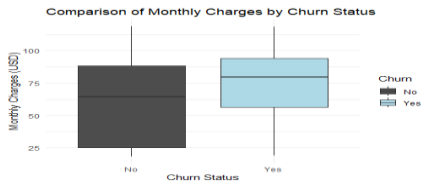


Figure 7. Comparison of Monthly Charges by Churn Status

Churned customers (Yes) tend to have higher median Monthly Charges than non-churned customers (No). The box plot for churned customers is noticeably shifted upwards, indicating they generally pay more per month.

02.Tenure

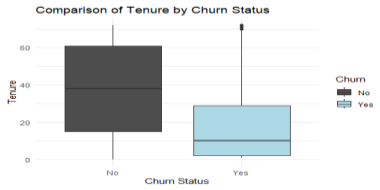
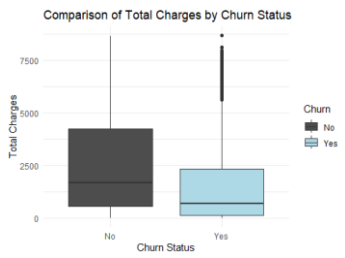


Figure 8. Comparison of Tenure by Churn Status

Churned customers (Yes) have a significantly lower median Tenure (number of months with the company) than non-churned customers (No). This suggests that customers who leave tend to be newer customers.

03.Total Charges

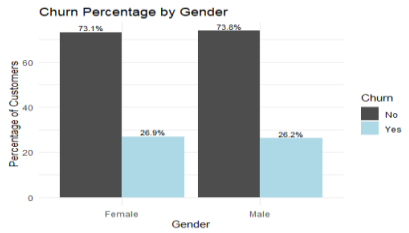


Churned customers (Yes) have a significantly lower median Total Charges than non-churned customers (No). This is consistent with their lower tenure, as they have accumulated fewer total charges.

Figure 9. Comparison of Total Charges by Churn Status

Categorical Variable

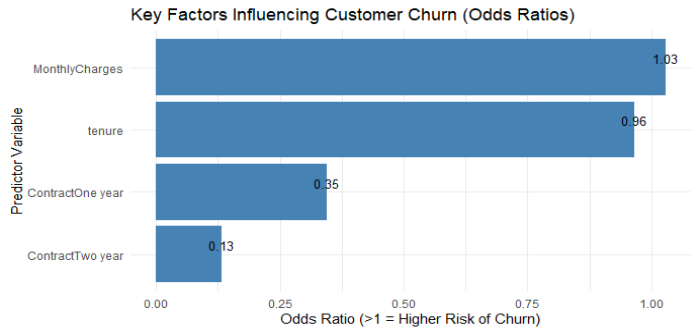
01.Gender vs Churn



The churn rate is very similar for both female (26.9%) and male (26.2%) customers. This indicates that **Gender is not a key factor** influencing customer churn.

Figure 10. Churn Percentage by Gender

OBJECTIVE 3: To provide practical recommendations based on key churn factors.



Monthly Charges (Odds Ratio = **1.03**) are positively related to churn, meaning higher monthly charges **increase the risk of churn**. Conversely, **Tenure** (0.96), **One-year Contract** (0.35), and **Two-year Contract** (0.13) all have Odds Ratios less than 1.0, indicating that longer tenure and having a fixed-term contract **significantly decrease the risk of churn**. Having a two-year contract provides the greatest reduction in churn risk.

Figure 11. Key Factors Influencing Customer Churn (Odds Ratios)

OBJECTIVE 4: To develop a statistical model that predicts the likelihood of churn.

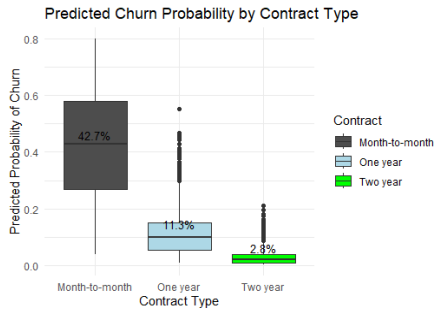


Figure 12. Predicted Churn Probability by Contract Type

The statistical model (Logistic Regression) shows that **Contract Type is the single most influential predictor of churn risk**.

- **High Risk (Month-to-month):** Customers with a Month-to-month contract have the highest risk, with a median predicted churn probability of **42.7%**.
- **Low Risk (Fixed Term):** Churn probability drops dramatically for fixed-term contracts: **11.3%** for One-year contracts and just **2.8%** for Two-year contracts.

This confirms the model strongly associates **longer contracts with much lower churn likelihood**.

General Discussion and Conclusion

- The dominant driver of churn is **contract type**. While month-to-month customers have the highest rate of churn, 42.7%, this rate is substantially lower for two-year contracts, at 2.8%. Therefore, longer contracts can promote lower rates of churning.
- The **type of service** does matter. Fiber-optic customers have higher churn (≈41.9%) than either DSL or no-internet customers; this may reflect service dissatisfaction or price sensitivity for fiber subscribers. Explore further service quality and price against this segment.
- **Tenure and financial profile:** Churners have much lower tenure and much higher median monthly charges - most of them leave early and pay more per month. That means early intervention during the first months can be effective.
- **Demographic signals:** senior citizen and customers with no partners/dependents have a greater tendency to churn and thus target specific segments for retention campaigns.
- **Model insights and predictive utility:** Logistic regression confirms the descriptive patterns that higher monthly charges increase churn risk (OR ≈1.03 per USD), and fixed-term contracts strongly reduce risk-one-year OR ≈ 0.35, two-year OR ≈ 0.13. The model performance represents a well-performing model that can be used to prioritize outreach, but its implementation should be complemented by checks for calibration.
- **Practical recommendations:** Retention programs month-to-month and fiber; Incentives to convert high-risk customers to fixed-term contracts including discounts and bundles; Proactive outreach deployed months 1–3 including welcome calls and early discounts; Targeted CRM program to be run for senior customers and high subscribers of a monthly charge.